Geochemistry Geophysics Geosystems G3
AN ELECTRONIC JOURNAL OF THE EARTH SCIENCES
Published by AGU and the Geochemical Society

Article
Volume 5, Number 3
12 March 2004
Q03H09, doi:10.1029/2003GC000632
ISSN: 1525-2027

Paleomagnetic results from the Newer Volcanics of Victoria: Contribution to the Time Averaged Field Initiative

Neil D. Opdyke
Department of Geological Sciences, University of Florida, P.O. Box 112120, Gainesville, Florida 32611, USA (drno@ufl.edu)

Robert Musgrave
Department of Earth Sciences, La Trobe University, Bundoora, Victoria, 3086 Australia (r.musgrave@latrobe.edu.au)

[1] The Newer Volcanics of Victoria Australia were sampled in 2001 and 2003 as part of the Time Averaged Field Initiative which is designed to provide a reliable data set for modeling the Earth's magnetic field over the past 5 My. The collection consisted of ten cores per site at 42 sites. Complete demagnetization was carried out on all samples using AF or thermal demagnetization and line fitting techniques were employed to determine the resultant vector. Thirty-two of these sites gave results that met the criterion that $N \geq 5$, and α_{95} was less than 6°. 20 sites were reversely magnetized, Dec = 173.4°, Inc. = 54.6°, α_{95} = 3.8° and 13 normally magnetized, Dec = 2.1°, Inc = −62.3°, α_{95} = 6.3°. The data gave a global mean paleomagnetic pole (lat. 87.2°, long. 21.9° and A_{95} = 4.4°) which is not significantly different from the pole of rotation for the present location of Victoria and for its latitude 2.5 million years ago when most of the lavas were erupted.

Components: 3952 words, 7 figures, 2 tables.

Keywords: Time-averaged field; paleomagnetic secular variation; Newer Volcanics; geocentric axial dipole; visual geomagnetic pole.

Index Terms: 1522 Geomagnetism and Paleomagnetism: Paleomagnetic secular variation; 1560 Geomagnetism and Paleomagnetism: Time variations—secular and long term; 1530 Geomagnetism and Paleomagnetism: Rapid time variations.

Received 11 September 2003; **Revised** 14 January 2004; **Accepted** 20 January 2004; **Published** 12 March 2004.

Opdyke, N. D., and R. Musgrave (2004), Paleomagnetic results from the Newer Volcanics of Victoria: Contribution to the Time Averaged Field Initiative, *Geochem. Geophys. Geosyst.*, *5*, Q03H09, doi:10.1029/2003GC000632.

Theme: Geomagnetic Field Behavior Over the Past 5 Myr **Guest Editors:** Cathy Constable and Catherine Johnson

1. Introduction

[2] The Newer Volcanics of Australia have been a favorite target for studies of paleosecular variation of the earth's magnetic field. They were first studied by *Irving and Green* [1957], using the NRM vector only. There was no significant difference between the directions observed and that expected from a geocentric axial dipole field. This was followed by a study by *Aziz-ur-Rahman* [1971] who used blanket demagnetization and obtained a result similar to Irving and Green's study. The work reported here uses modern demagnetization techniques in an attempt to update these previous studies. This study is part of the Time Averaged Field Initiative (TAFI) whose pur-

Copyright 2004 by the American Geophysical Union

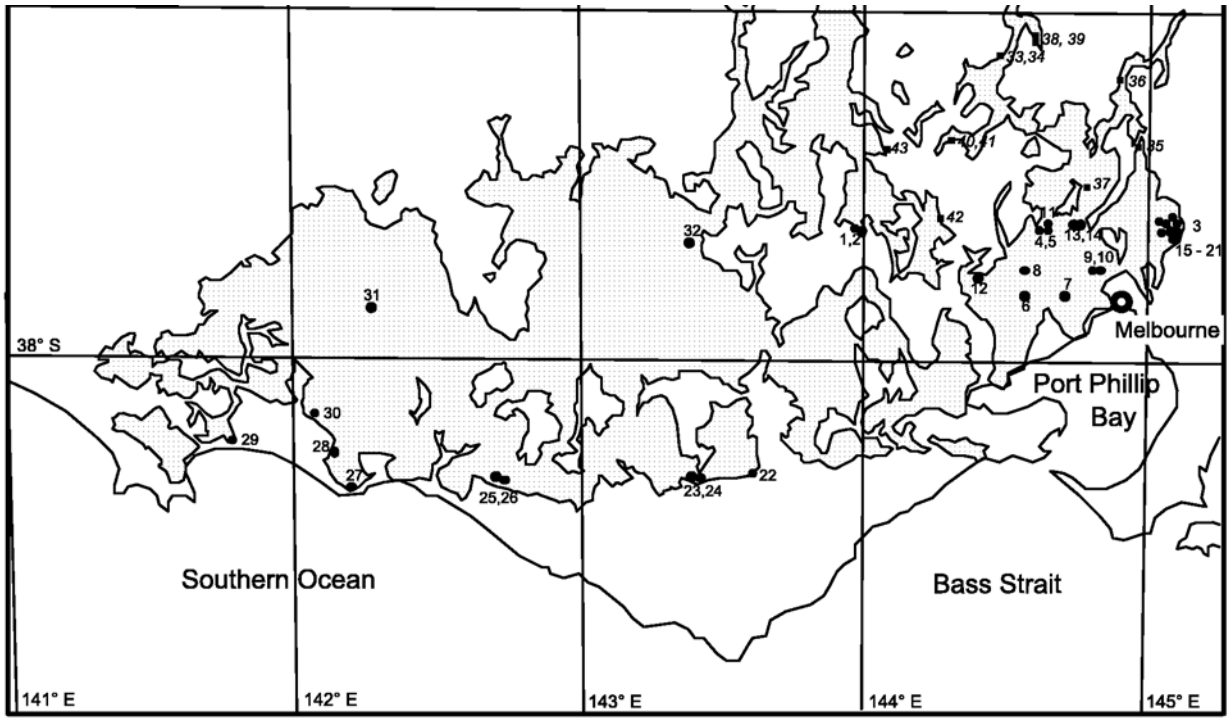


Figure 1. Outcrop map of Newer Volcanics with site locations of the sampling localities indicated by filled circles (2001 collection) and filled squares (2003 collection).

pose is to update and improve the available data in order to model the Earth's magnetic field in the past. Results from Australia will serve to fill a gap in the coverage of the southern hemisphere. Reversal transition paths seem to favor a route that passes over Australia making it important to know if the long term behavior of the field in the region is in any way unique [*Hoffman*, 1991].

2. Geology

[3] The Newer Volcanics of Victoria crop out over an area of 15000 km^2, extending in a broad east-west band from the western part of the urban area of Melbourne to the southeastern corner of South Australia (Figure 1). This suite of alkalic to tholeiitic basalts overlies Paleozoic basement of the Lachlan and Adelaide fold belts and the Mesozoic to Tertiary syn- and post-rift sediments of the Otway Basin. Two geomorphological provinces have been defined within the extent of the Newer Volcanics [*Joyce*, 1988]: the Highlands province in the northeast, characterized by numerous small eruption centers and valley-confined flows, and the Plains province which stretches across the southern extent of the basalts, and comprises small central vent features (scoria cones, maars, and low shields) surrounded by elliptical flow aprons. Individual flows extend up to about 100 km in length, and are typically less than 10 m thick; the net thickness of the Newer Volcanics sequence over the Plains province is commonly about 50 m or less. A total of more than 400 eruption points results in continuous flow converge over much of the extent of the Newer Volcanics.

[4] Compositions of basalts from the eruption features are commonly alkalic (alkali olivine basalts, hawaiities, and nepheline basanites and mugearites), and range through to olivine tholeiites. Flows of the Plains province, which formed the bulk of our sampling, are dominantly tholeiitic [*Price et al.*, 1988]. Strontium isotopic and trace element geochemistry has allowed the definition of a number of compositional domains, inferred by *Price et al.* [1997] to result form local geochemical heterogeneneity in the lithospheric mantle, super-

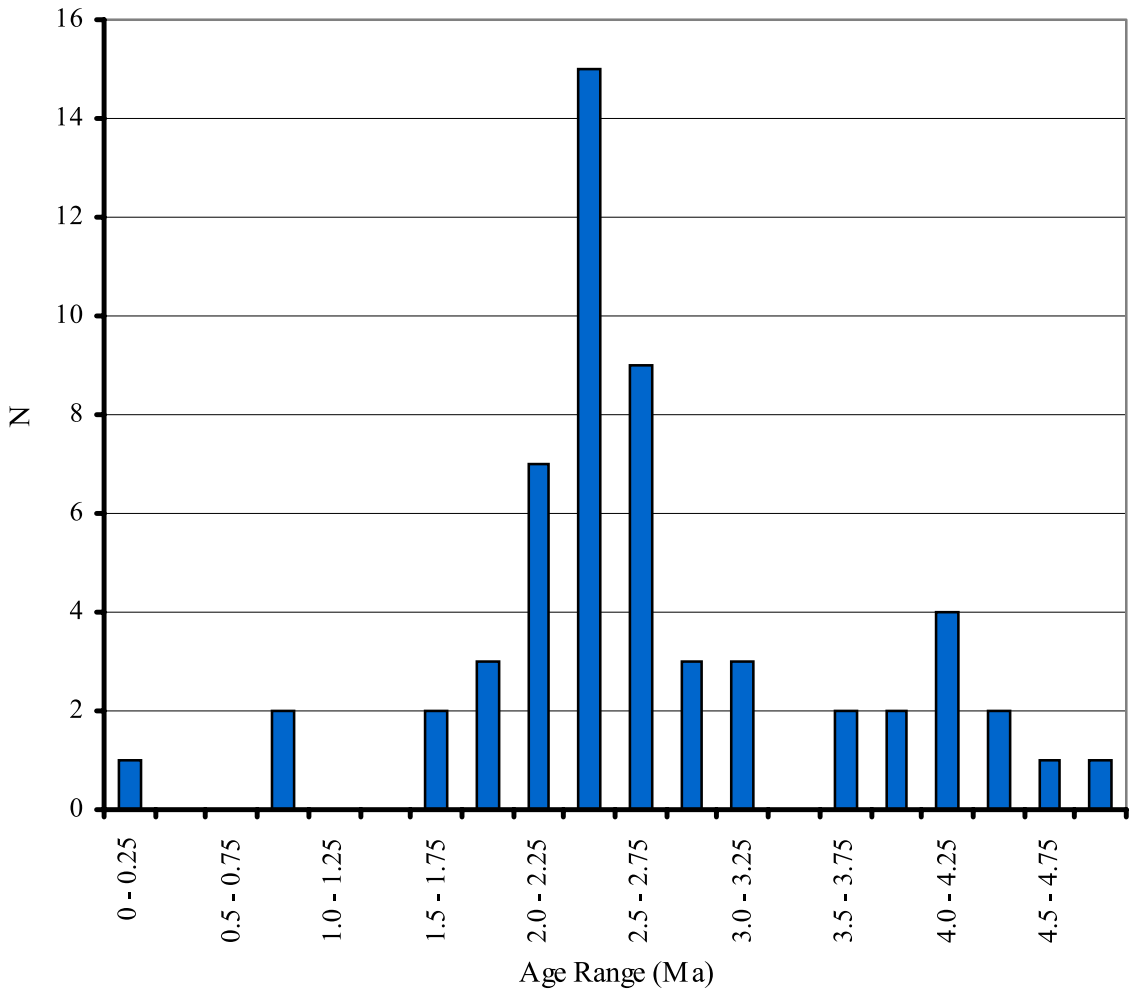


Figure 2. Frequency of occurrence of K/Ar ages in the Newer Volcanics.

imposed on a deeper plume source [*McDonough et al.*, 1985].

3. Dating

[5] The age of the Newer Volcanics ranges from 4 Ma to as young as 7000 years. The volcanicity seems to have reached a peak in the Pliocene in late Gauss and early Matuyama time [*McDougall et al.*, 1966; *Aziz-ur-Rahman*, 1972; *Singleton et al.*, 1976; *Wellman*, 1974]. The available radiometric dates are K-Ar dates and although many are old they are unlikely to have large errors. Figure 2 is a histogram of all available dates from the Newer Volcanics. The dates range between 0.25 and 5 Ma. The principle mode falls between 2.25 and 2.5 Ma with 42 (74%) falling between 1.5 and 3.25 Ma.

4. Sampling

[6] Samples were collected in the field using a hand held drill and oriented using a magnetic compass and where possible a sun compass. Ten independently oriented cores were taken from every site and the site location was determined by GPS. We attempted to sample over the outcrop area of the Newer Volcanics. The sampling localities were usually quarries, road cuts or river beds chosen to minimize exposure to lightening strikes. The site distribution is shown on Figure 1. It can be

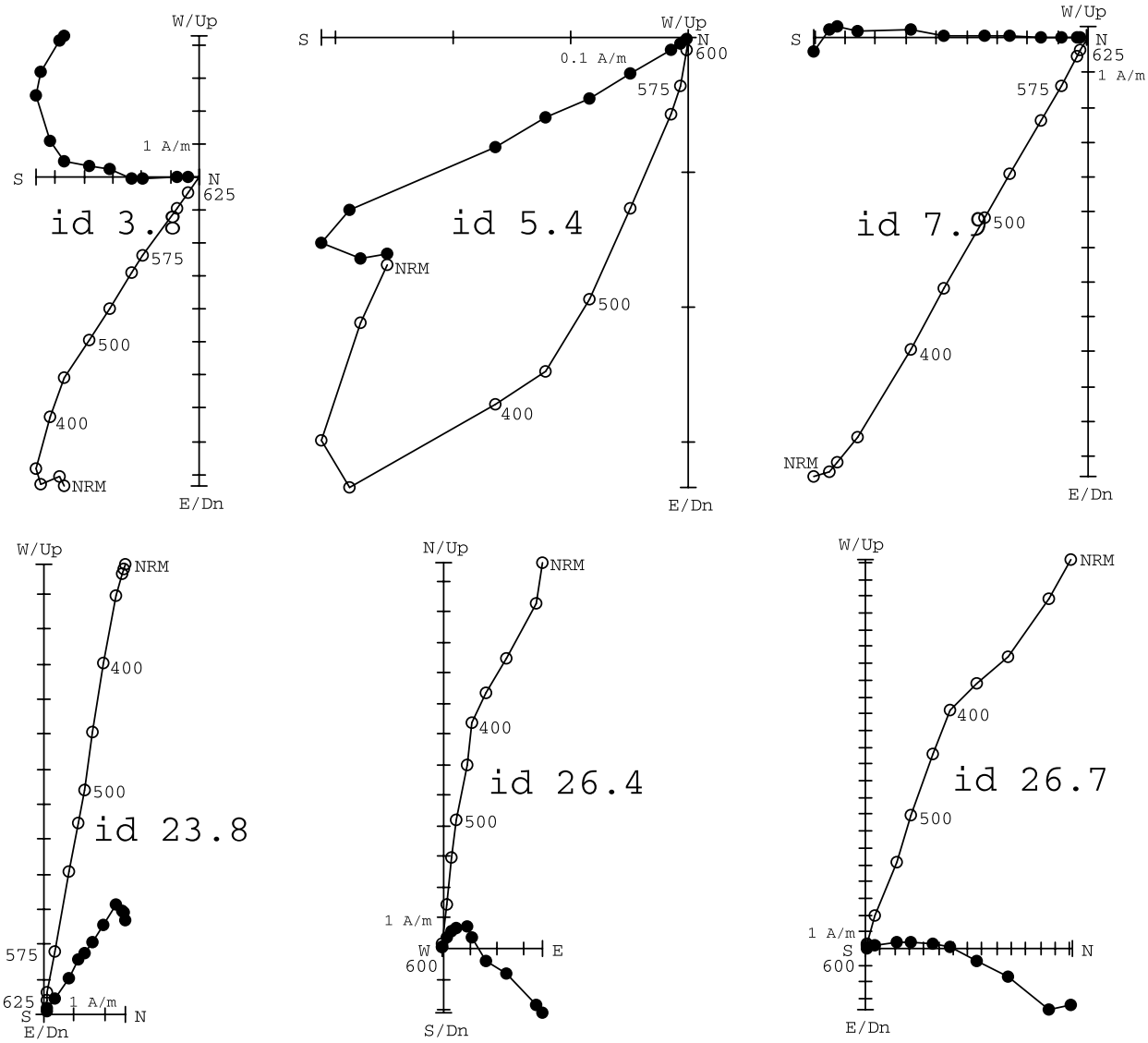


Figure 3. Demagnetograms of samples from the 2001 collection. Filled circles represent the end points of the horizontal component while the open circles represent the end points of the vertical component.

seen that the sites are well distributed over the area of outcrop of the Newer Volcanics although there is a higher density of sampling in the vicinity of Melbourne. Samples were collected in 2001 by the authors working together and ten additional sites were collected by Musgrave in 2003.

5. Laboratory Studies

[7] The cores were returned to the laboratory where they were sliced into specimens and were measured in the paleomagnetic laboratory at the University of Florida. The measurements were made in a 2G cryogenic magnetometer housed in a magnetically shielded room. One sample from each site was demagnetized using alternating fields with the rest of the samples demagnetized using thermal or AF demagnetization. Some sites were affected by lightning and were cleaned using alternating fields. The directions of magnetization were determined by line fitting techniques [*Kirschvink*, 1980] and combined using *Fisher*

Table 1. Site Statistics[a]

Site	Lon °E	Lat °S	n/N	Dec, deg	Inc, deg	α_{95}	κ	P	Lon °E	Lat °N	A_{95}
1	145.0	37.7	10/10	163.0	49.2	1.8	686.8	R	258.4	−74.0	2.0
2	145.0	37.7	8/10	161.2	46.8	3.3	289.0	R	261.1	−71.3	3.4
3	145.1	37.7	8/10	173.7	46.5	4.9	126.7	R	295.2	−78.8	5.0
4	144.6	37.7	8/10	184.6	60.7	4.5	149.6	R	104.6	−84.7	6.0
5	144.7	37.6	9/10	169.3	59.6	5.8	79.4	R	212.5	−81.2	7.6
6	144.6	37.8	7/10	183.6	59.3	6.3	92.9	R	95.1	−86.4	8.2
7	144.7	37.8	9/10	171.9	58.7	4.1	158.1	R	217.7	−83.5	5.2
8	144.6	37.7	10/10	181.1	53.4	3.5	187.9	R	338.3	−86.1	4.1
9	144.8	37.7	7/10	27.6	−64.6	1.5	1626.0	N	267.0	67.9	2.1
10	144.8	37.7	7/10	5.9	−55.8	2.2	785.4	N	220.5	85.1	2.7
11	144.7	37.7	9/10	179.1	55.4	4.4	139.1	R	302.2	−88.1	5.3
12	144.4	37.7	10/10	4.7	−63.0	1.8	685.6	N	298.3	82.4	2.5
13	144.8	37.6	10/10	353.3	−55.4	4.4	120.1	N	70.0	84.4	5.3
14	144.8	37.6	10/10	13.7	−62.4	2.1	525.0	N	269.7	78.0	2.9
15	145.1	37.6	8/10	170.5	50.6	2.5	493.3	R	271.0	−80.0	3.0
16	145.0	37.6	28/30	168.7	48.6	1.4	372.1	R	272.2	−77.6	1.5
19	145.0	37.6	10/10	174.2	50.1	2.6	358.5	R	287.8	−81.7	2.8
20	145.0	37.5	16/20	169.2	47.8	1.9	370.8	R	275.6	−77.5	2.0
22	143.6	38.3	9/10	7.3	−53.4	4.7	118.8	N	199.4	82.7	5.4
23	143.4	38.3	18/20	294.7	−56.0	2.6	184.6	N	33.8	39.2	3.2
25	142.7	38.2	9/10	359.1	−71.2	5.9	78.1	N	324.4	72.4	9.6
26	142.7	38.2	9/10	6.0	−72.1	3.8	187.2	N	312.8	70.7	6.3
27	142.2	38.4	9/10	327.0	−39.7	5.3	95.3	N	72.0	57.7	4.9
29	141.8	38.2	10/10	17.3	−69.7	4.2	132.5	N	289.6	70.6	6.7
30	142.1	38.2	6/10	352.9	−65.5	2.9	531.8	N	348.5	79.2	4.2
31	142.3	37.9	9/10	333.2	−45.2	12.2	18.8	N	70.8	64.9	12.3
32	143.4	37.7	10/10	258.0	−64.9	2.0	601.3	N	8.6	19.5	2.9
33	144.5	37.1	10/10	176.9	64.2	2.0	572.9	R	158.3	−80.9	2.9
34	144.5	37.1	9/10	167.4	46.3	3.9	175.3	R	272.6	−75.8	4.0
35	145.0	37.4	9/10	184.3	42.1	4.8	115.9	R	341.7	−76.4	4.6
36	144.9	37.2	10/10	185.1	52.1	3.2	224.9	R	9.6	−83.9	3.6
37	144.8	37.5	10/10	29.0	−61.6	2.7	323.7	N	257.2	67.3	3.6
38	144.6	37.0	10/10	170.2	64.0	1.6	936.2	R	181.5	−78.6	2.2
39	144.6	37.0	10/10	190.2	63.3	2.5	377.8	R	103.2	−79.1	3.5
40	144.3	37.4	5/5	350.2	−50.4	8.0	91.6	N	89.3	79.8	8.8
41	144.3	37.4	5/5	189.9	58.6	7.9	93.7	R	71.5	−82.0	10.2
42	144.3	37.6	10/10	349.3	−60.6	2.2	503.2	N	25.4	80.9	2.9
43	144.1	37.4	10/10	141.9	65.6	3.5	193.9	R	201.4	−60.4	5.1

[a] n/N: number of samples used/measured, Dec: declination, Inc: inclination, P: polarity, κ: precision parameter, α_{95} (A_{95}): 95% confidence circles about directions (VGPs).

[1953] statistics. MAD values were small and in most cases five or more data points were used to determine the resultant vector however in some instances because of curved demagnetization trajectories caused by lightening, four data points were used to determine the resultant vector (Figure 3 sample 3.8).

[8] The site statistics are presented in Table 1. The study consists of a collection of 43 sites; however, it became clear after the data were obtained that in three cases the same flow was sampled more than once; therefore sites 16, 17 and 18 were combined into one site designated site 16; sites 20 and 21 were combined as site 20 and sites 23 and 24 combined as site 23. One site yielded 5 normally magnetized and 5 reversely magnetized samples, apparently two flows were sampled at this locality and are presented in Table 1 as sites 40 and 41. Site 28 did not respond to either AF or thermal demagnetization and the results are highly scattered from this site and are not presented in Table 1. At other sites samples that yielded directions away from the main grouping were not included in the site statistics. These samples may have been affected by lightning or in some cases portions of the outcrop

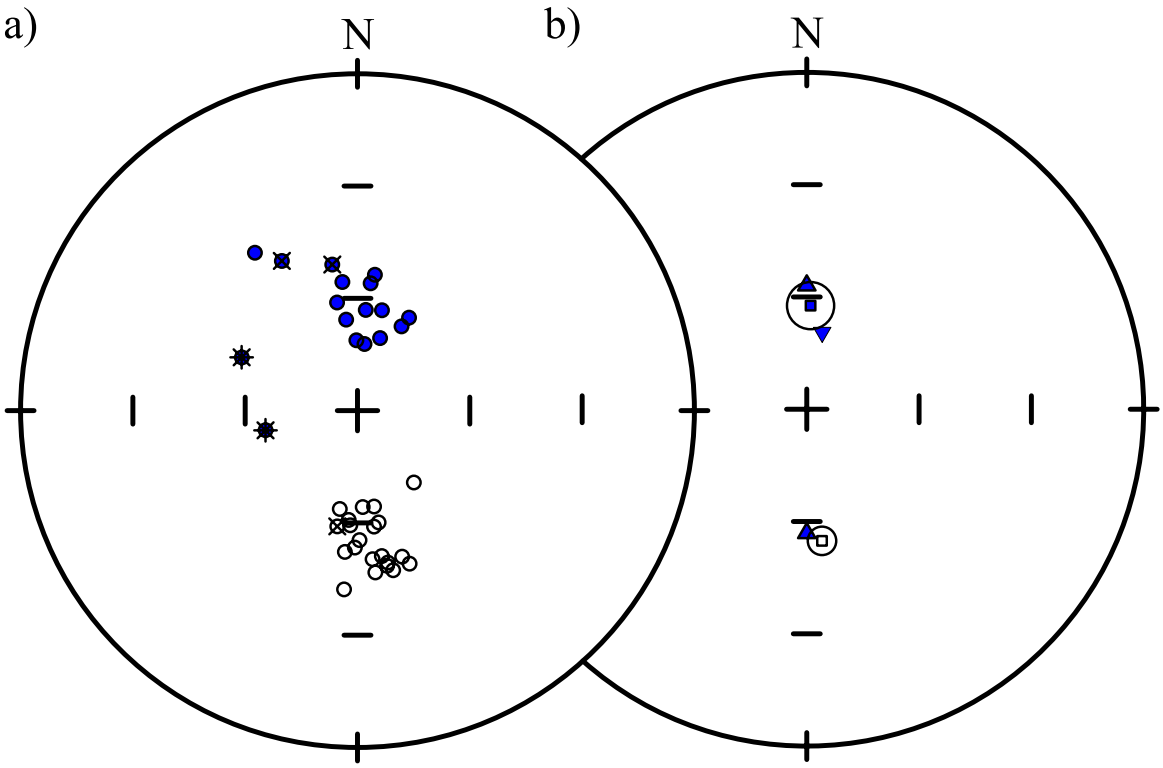


Figure 4. Directions of magnetization plotted on a stereographic projection. Directions above (below) the horizontal are represented by filled (open) circles. Figure 4a shows site mean directions. Sites not used in further calculations are either crossed (not well grouped) or marked with an asterisk (transitional). Figure 4b shows the mean direction of the normal sites (filled square) and reverse sites (open square) with their 95% confidence limits (circles). Filled triangles pointing up are the GAD for normal and reverse fields. Filled triangle pointing down is the 2001 IGRF.

may have been subjected to unrecognized rotations or gross orientation error. The sites yield results that are well grouped except for site 31 which had an α_{95} of 12° and sites 40 and 41 which have α_{95} of 8°. Thirty five sites had α_{95} less than 6.3° and two sites (23, 32) are excluded from further computations since they yield VGPs at low latitudes and are interpreted as transitional directions. Thirteen of the sites gave normal directions of magnetization while 20 are reversed. Figure 4 shows the site mean directions plotted on a stereonet illustrating that the reverse and normal sites are well grouped except for sites 23 and 32 which appear to be associated with a reversal or an excursion of the Earth's magnetic field. The VGPs calculated from these studies are shown in Figure 5a with sites 23 and 32 excluded since the VGPs from these sites fall at latitudes below 40°. The reverse and normal site mean directions were calculated excluding these sites and the results are given in Table 2. This analysis shows that the normal and reversed site means are significantly different from each other, with the normal sites being steeper than the dipole field and the reversed sites being shallower than the Earth's dipole field. The mean of the normal and reversed directions is not significantly different from the geocentric axial dipole.

[9] In most studies of PSVL corrections for plate tectonic motion are not made due to the fact that the data come from slow moving plates or the plates are moving along lines of paleolatitude. Australia, however, is on a fast moving plate moving north at 66 mm per year. This would amount to a movement to the north of 66 Kilometers every million years. The radiometric dating shows that most lava flows were extruded at about

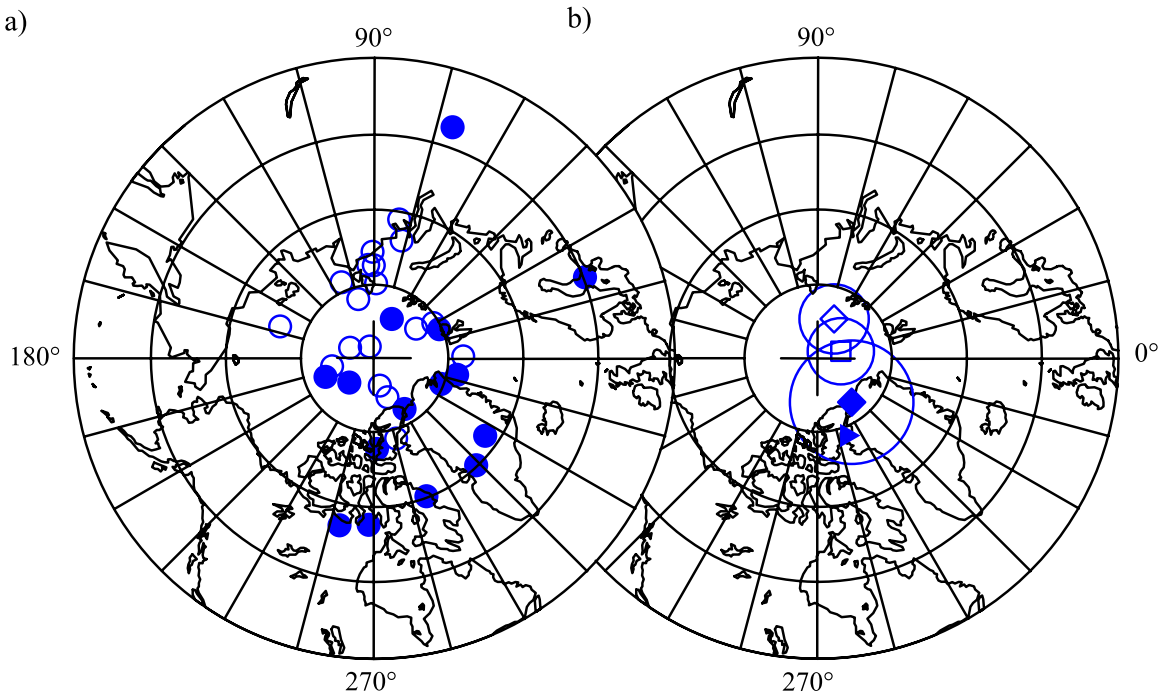


Figure 5. Dispersion of VGPs. Filled symbols represent normal directions of magnetization while the reversed sites are indicated by open symbols. Figure 5a shows the site mean VGPS and Figure 5b shows the paleomagnetic poles for reversed (open diamond) and normal (filled diamond) data sets. The open square is the overall mean paleomagnetic pole. The filled triangle shows the position of the geomagnetic North pole.

2.5 million years ago when Victoria was at a paleolatitude of 39.5° south which would predict an inclination of about 58.76° which can be compared with the observed value of 58.1°. The mean of the reversed and normal sites are combined and the resulting paleomagnetic pole is again not significantly different from the present geocentric axial dipole (Figure 5b) however it agrees best with the dipole field corrected for plate motion.

6. Discussion

[10] Studies of the Time Averaged Field rest on the assumption that each site is an observation of the

Table 2. Statistics of Combined Sites[a]

	Dec, deg	Inc, deg	α_{95}	N	κ	P long	P lat	A_{95}	K	OG	OA	ASD
All w/o Trans N	358.3	−60.6	6.3	15	38	326.8	85.1	8.1	23	yes	yes	17.0
All w/o Trans R	174.1	54.9	3.7	21	76	252.3	−85	4.5	51	no	no	12.1
All w/o Trans com	355.7	−57.3	3.4	36	51	33.5	86.9	4.2	33	yes	yes	14.4
<6 N	2.1	−62.3	6.3	13	44	305.2	83	8.2	27	yes	yes	16.7
<6 R	173.4	54.6	3.8	20	77	252.3	−84.4	4.6	52	no	no	12.3
<6 com	356.3	−57.7	3.5	33	52	21.9	87.2	4.4	33	yes	yes	14.2
<5 N	2.2	−61.5	6.7	12	42	300.3	83.8	8.7	26	yes	yes	16.6
<5 R	173.1	54	4.1	18	72	255.7	−83.9	5	49	no	no	12.8
<5 com	356.3	−57.1	3.8	30	50	34.1	87.4	4.7	32	yes	yes	14.4

[a] Statistical results by site groups, calculated using *Fisher* [1953] statistics. Groups comprise all sites except transitional data (all w/o trans) for Normal, N, Reversed, R and Combined data, as well as the same combinations for groups of sites with α_{95} less than 6 and 5. In addition other abbreviations are as in Table 1, O.G./O.A indicates whether the 95% confidence limits (α_{95}) of the mean direction/mean VGP overlap the GAD/Earth's rotation axis and ASD is the angular standard deviation (degrees) of the VGPs relative to the Earth's axis of rotation.

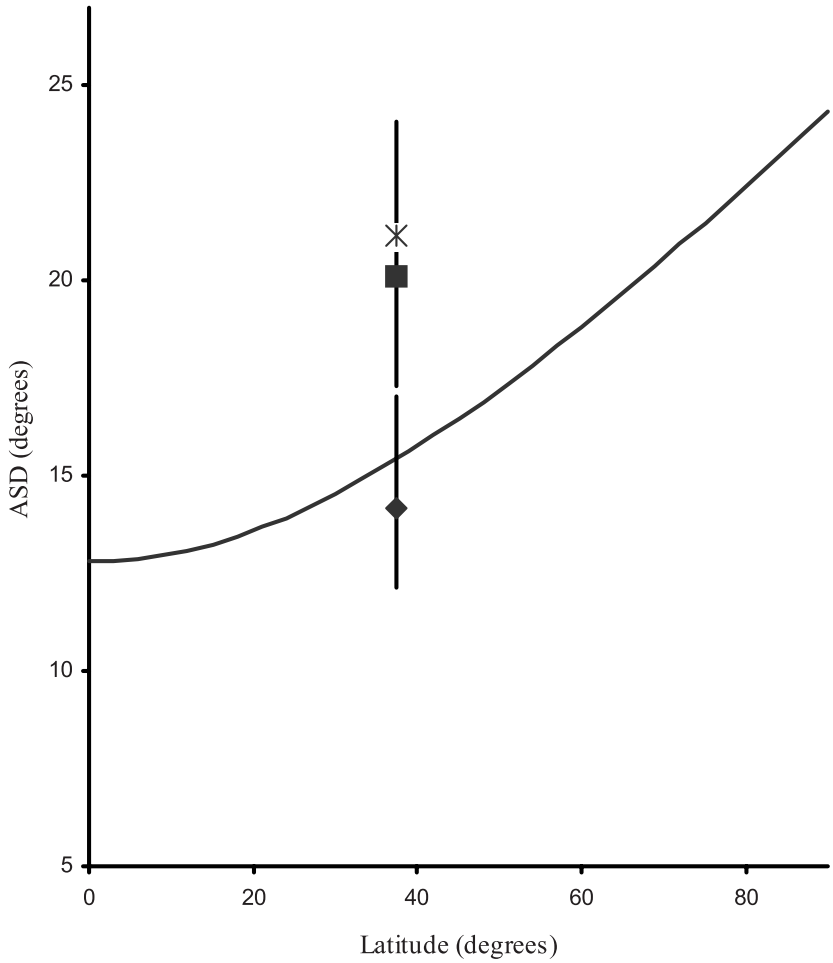


Figure 6. VGP scatter relative to the Earth's axis of rotation. The figure shows: Model G (solid line), VGP scatter from this study excluding transitional sites (diamond), VGP scatter from this study including transition sites (square) and VGP scatter from the present field at 38°S (asterisk).

direction and in some cases the intensity of the magnetic field over a period of time on the order a week to a year depending on the thickness of the flow. These observations are necessarily compared to the parameters of the present Earth's magnetic field. The scatter of the observations can be displayed as variations of direction [*Creer*, 1962] or VGPs can be calculated from the directions and the dispersion of the VGPs around the geographic pole or the geomagnetic pole can be calculated. This method supposes that at any latitude the dispersion of the field around the Earth will be representative of the dispersion of the field in the past. *Cox* [1962] was the first to calculate the VGP dispersion relative to the geomagnetic pole: however, most recent authors have chosen to measure the dispersion relative to the geographic pole. The present magnetic field is asymmetric about the equator with a much higher dispersion of the field in the southern hemisphere than in the northern hemisphere, [*Merrill et al.*, 1996; *Tauxe*, 1998] with values of angular standard deviation reaching 30° at high latitudes in the southern hemisphere while in the northern hemisphere at high latitudes the dispersion is ten degrees or less. The dispersion of the present field when averaged between the northern and southern hemispheres is remarkably similar to model G of *McFadden et al.* [1991]. The directional data from the sites in this study were used to calculate VGPs from each site (Figure 5), The dispersion of the VGPs (with the exception of the transitional sites) about the axis of rotation

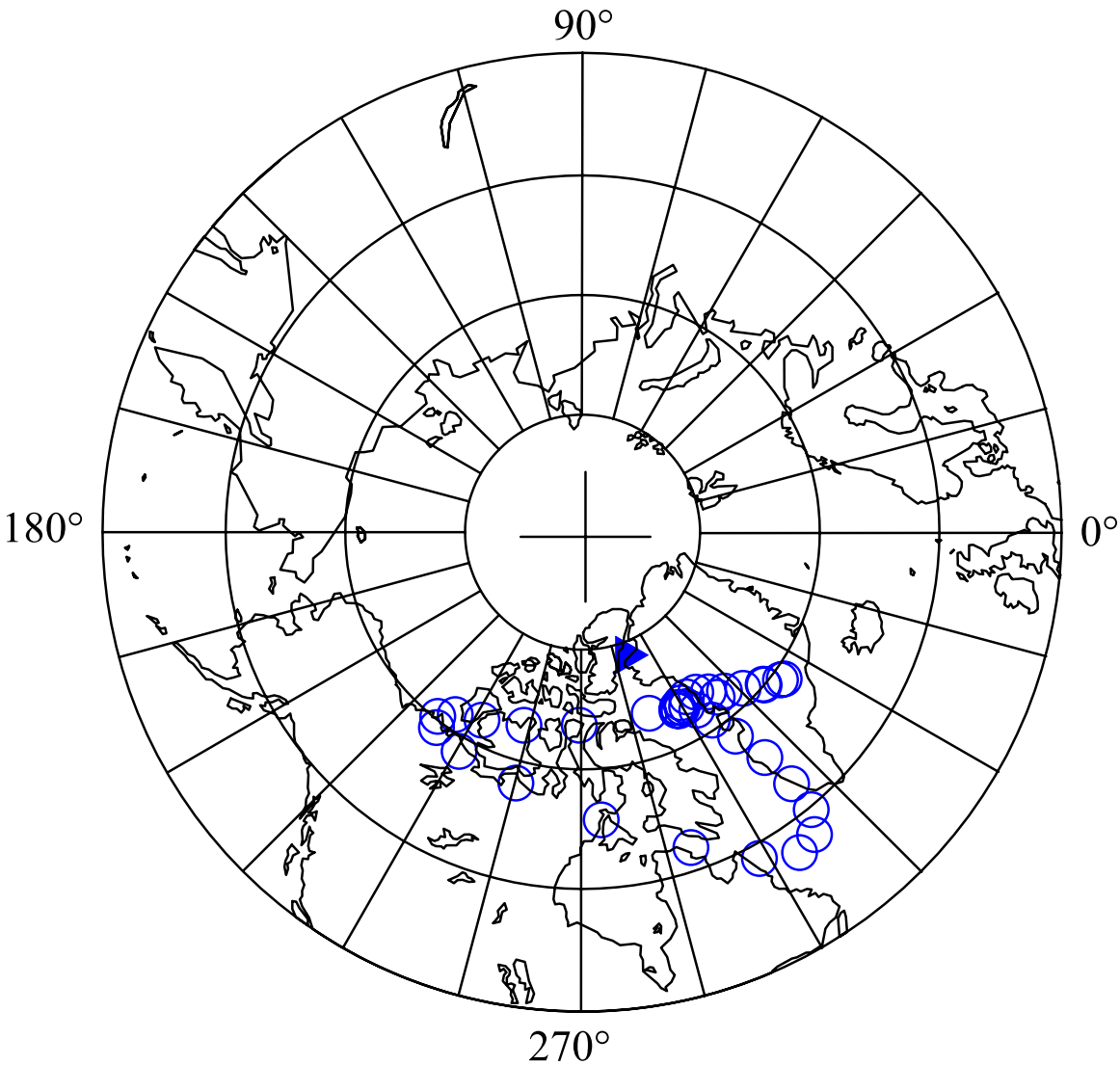


Figure 7. VGPs plotted for the present field at ten degrees intervals around the Earth at 38 degrees south latitude. The filled triangle represents the geomagnetic pole.

gives a value of angular standard deviation of 14.3° (with lower and upper 95% confidence limits of 12.1° and 17.0° respectively [*Cox*, 1969]) which is less than the value (15.7°) expected from model G of *McFadden et al.* [1991], (Figure 6) but not significantly so.

[11] In order to compare these results with the present Earth's magnetic field, data were taken from the latitude of the site (38°S) at ten degrees intervals around the Earth and VGPs plotted with respect to the axis of rotation (Figure 7). The dispersion for this data set is 21° which is much higher than dispersion of the data from the Newer Volcanics averaged over several million years. It is surprising that the field in the past has such a low VGP dispersion. If the present field is a reliable guide to past fields then larger dispersions would be expected based on the present field even though the average is in line with results predicted by model G. Two sites in this study are believed to be transitional. If the dispersion is recalculated including these two sites then the dispersion increases to 20.1°, much closer to the value given by the present field (21°). It would be expected that random samplings of the field would record directions as dispersed as those observed today. The data set that is observed over the last 5 Ma yields a

VGP scatter that ranges from 13° to about 20°, increasing with latitude. This would not be predicted on the basis of the present magnetic field of Earth. The present Earth's magnetic field therefore seems to be anomalous with respect to the Time Averaged Field. It has been suggested by *Hulot et al.* [2002] that Earth's field is becoming a transitional field and is in the process of reversing polarity. This hypothesis has been put forward in the past to explain the decrease in intensity of the field observed over the last 100 years; however, *Hulot et al.* [2002] put forward this suggestion to explain the rapid change in Earth's field over the last 20 years. The Time Averaged Field data would seem to support this hypothesis. It would seem that for most of the time Earth's field has been much more stable than it is today.

[12] The data presented here is dominated by the geocentric axial dipole (GAD) and the mean paleomagnetic pole is not significantly different from the present axis of rotation. This observation is similar to other recent studies that have not observed a far sided right handed effect as might be expected from earlier studies [*Wilson*, 1972; *Merrill et al.*, 1996]. It may be that this effect might be caused by unremoved overprints since it is more pronounced in reversed rocks [*Merrill et al.*, 1996].

Acknowledgments

[13] The authors would like to acknowledge Kainian Huang for his careful laboratory work and Victoria Mejia who helped with some of the computations and preparation of figures and many helpful discussions. The study was conducted with the support of NSF Grant EAR-9804737.

References

Aziz-ur-Rahman (1971), Paleomagnetic secular variation for recent normal and reversed epochs, from the Newer Volcanics of Victoria, Australia, *Geophys. J. R. Astron. Soc.*, *24*, 255–269.

Aziz-ur-Rahman (1972), Potassium-Argon ages on the Newer Volcanics of Victoria, *Proc. R. Soc. Victoria*, *85*, 61–70.

Cox, A. (1962), Analysis of the present Geomagnetic field for comparison with paleomagnetic results, *J. Geomag. Geoelect.*, *13*, 101–112.

Cox, A. (1969), Confidence limits of the precision parameter k, *Geophys. J. R. Astron. Soc.*, *18*, 545–549.

Creer, K. M. (1962), The dispersion of the geomagnetic field due to secular variation and its determination for remote times from paleomagnetic data, *J. Geophys. Res.*, *67*, 3461–3476.

Fisher, R. A. (1953), Dispersion on a sphere, *Proc. R. Soc. London*, *A217*, 295–305.

Hoffman, K. A. (1991), Long-lived transitional states of the geomagnetic field and the two dynamo families, *Nature*, *354*, 273–277.

Hulot, G., C. Eymin, B. Langlais, M. Mandea, and N. Olsen (2002), Small-scale structure of the geodynamo inferred from Oersted and Magsat satellite data, *Nature*, *6881*, 620–623.

Irving, E., and R. Green (1957), Paleomagnetic evidence from the Cretaceous and Cainozoic, *Nature*, *179*, 1064–1065.

Joyce, E. B. (1988), Newer Volcanic landforms, in *Geology of Victoria*, edited by J. G. Douglas and J. A. Ferguson, pp. 419–426, Victorian Division Geological Soc. of Australia.

Kirschvink, J. L. (1980), The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718.

McDonough, W. F., M. T. McCulloch, and S. S. Sun (1985), Isotopic and geochemical systematics in Tertiary-Recent basalts from southeastern Australia and implications for the evolution of the sub-continental lithosphere, *Geochem. Cosmochem. Acta*, *49*, 2051–2067.

McDougall, I., H. I. Allsop, and F. H. Chamalaun (1966), Isotopic dating of the Newer Volcanics of Victoria, Australia, and geomagnetic polarity epochs, *J. Geophys. Res.*, *71*, 6107–6118.

McFadden, P. L., R. T. Merrill, M. W. McElhinny, and S. Lee (1991), Reversals of the Earth's Magnetic field and temporal variations of the dynamo families, *J. Geophys. Res.*, *96*, 3923–3933.

Merrill, R. T., M. W. McElhinny, and P. L. McFadden (1996), *The Magnetic Field of the Earth*, 507 pp., Academic, San Diego, Calif.

Price, R. C., C. M. Gray, I. A. Nicholls, and A. Day (1988), Cenozoic volcanic rocks, in *Geology of Victoria*, edited by J. G. Douglas and J. A. Ferguson, pp. 439–451, Victorian Div. Geol. Soc. Aust., Melbourne.

Price, R. C., C. M. Gray, and F. A. Frey (1997), Strontium isotope and trace element heterogeneity in the plains basalts of the Newer Volcanic Province, Victoria, Australia, *Geochim. Cosmochem. Acta*, *61*, 171–192.

Singleton, O. P., I. McDougall, and C. W. Mallet (1976), The Pliocene-Pleistocene boundary in Southeastern Australia, *J. Geol. Soc. Aust.*, *23*, 299–311.

Tauxe, L. (1998), Paleomagnetic principles and practices, 299 pp., Kluwer Acad., Norwell, Mass.

Wellman, P. (1974), Potassium-Argon ages on the Cainozoic Volcanic rocks of Eastern Australia, *J. Geol. Soc. Aust.*, *21*, 359–376.

Wilson, R. L. (1972), Paleomagnetic differences between normal and reversed field sources, and the problem of far-sided and right-handed pole positions, *Geophys. J. R. Astron. Soc.*, *28*, 295–304.